

PROFESSIONAL INTERVIEW GUIDE

12 Technical & Behavioral Questions and Answers for IT Support & Infrastructure Roles

Q1: Can you walk us through the technical details of the IT infrastructure project you designed for Ubuntu Innovations?

I designed a production-ready IT framework for a 25-employee startup expanding into a new office. The solution spanned a comprehensive business needs analysis, hardware/software procurement, and a structured network topology. I mapped an ISP link down to a business-grade firewall router and a 24-port Gigabit managed switch. For secure data access, I implemented a segmented IP addressing plan alongside a granular Linux directory structure (`/Company/`) utilizing security groups to enforce least-privilege access.

Q2: How did you structure the network addressing plan for the 25 employees, and what was your business rationale?

I deployed a Class C private IP addressing scheme using a `/24 subnet mask` to segment network assets into distinct functional domains. Specifically, I assigned subnets for Servers/NAS (`192.168.10.0/24`), User Devices (`192.168.20.0/24`), Network Printers (`192.168.30.0/24`), and Corporate Wi-Fi (`192.168.40.0/24`). This segmentation isolates broadcast domains, optimizes performance, and protects core assets from lateral discovery or unauthorized Wi-Fi access.

Q3: If a user tells you their desktop computer won't power on, what are the exact troubleshooting steps you would follow?

I follow a systematic physical-layer elimination process. First, I check external connections—ensuring the power cable is firmly seated in the PSU, wall outlet, and UPS, and confirming switches are on. Second, I look for signs of life such as LEDs, fans, or POST beeps. If dead, I perform a power cycle by holding the power button for 15 seconds with the cable unplugged to drain residual electricity. Finally, I swap the power cable or socket to determine if the failure is external or an internal PSU fault.

Q4: Explain how you would set up a shared directory structure in a Linux environment where only the Finance department can access financial records, but everyone can access public files.

First, I would verify the path and create the directories using `mkdir -p /Company/Public /Company/Finance`. Next, I would create the department group with `groupadd finance_grp` and assign it to the folder using `chown :finance_grp /Company/Finance`. To restrict access, I would set permissions to `chmod 770` on the Finance folder, limiting access strictly to the owner and group members. For the open directory, I would apply `chmod 775` so all employees can safely read and access public files.

Q5: What strategy did you put in place to ensure business continuity and minimize data loss for the company?

I engineered a multi-tiered Backup and Disaster Recovery Plan. Onsite, the infrastructure captures daily incremental modifications, weekly full snapshots, and monthly archives, writing them directly to a local **8TB RAID NAS** to guard against immediate hardware failure. For complete disaster resilience, these backups are automatically duplicated to secure, offsite cloud storage, successfully meeting our target recovery objectives (RTO/RPO) and minimizing downtime.

Q6: A user reports that their device states 'No Internet Connectivity.' How do you diagnose and resolve this issue?

I diagnose using a bottom-up OSI approach. I verify physical link lights on the NIC and switchport. Moving to the network layer, I check the network configuration via CLI (**ipconfig** / **ip address**) to ensure it hasn't fallen back to an APIPA address (**169.254.x.x**), which indicates DHCP failure. If valid, I ping the loopback address (**127.0.0.1**), the default gateway to check the local network, and an external IP like **8.8.8.8** followed by a domain name to isolate whether it's a routing fault or a local DNS issue.

Q7: Why is it important to use Uninterruptible Power Supplies (UPS) in a business-grade infrastructure environment?

A UPS provides essential power conditioning and business continuity. First, it acts as a surge protector that cleans incoming electricity, shielding endpoints, switches, and storage arrays from harmful spikes and brownouts. Second, during blackouts or load-shedding, its battery capacity keeps core components online. This prevents file system corruption caused by abrupt, "dirty" shutdowns and allows sufficient time for systems to power down gracefully.

Q8: How would you explain to a non-technical employee why they must implement Multi-Factor Authentication (MFA) on their accounts?

I use a simple home-security analogy. A password alone is like locking your front door with a single key; if someone steals or copies it via a phishing link, they can walk right in. Enrolling in Multi-Factor Authentication (MFA) is like adding a second security gate or a biometric scanner to that door. Even if a bad actor manages to steal your password, they still cannot gain entry because they cannot replicate the unique, temporary security code sent directly to your physical smartphone.

Q9: What are the phases of an Incident Response Plan, and what is your priority during a malware infection?

An Incident Response Plan flows through six stages: **Detection, Reporting, Containment, Eradication, Recovery, and Lessons Learned**. During an active malware infection, my absolute operational priority is containment. I immediately sever the infected endpoint's connection by pulling its Ethernet cable and disabling Wi-Fi. This stops the compromise in its tracks, preventing the malware from spreading laterally to other systems or accessing shared network databases.

Q10: How do you manage your daily IT support tasks when working inside an Agile or Scrum environment?

Within a Scrum framework, tasks are managed and prioritized in a sprint backlog based on impact and severity. I start my day at the Daily Standup to review progress, coordinate tasks, and raise any technical blockers. If an emergency issue breaks out mid-sprint, I log the ticket, coordinate briefly with the Scrum Master to evaluate its urgency against current commitments, and adjust my workflow to clear high-impact infrastructure obstacles without derailing sprint goals.

Q11: A network printer is showing as 'Offline' for multiple staff members. What is your diagnostic approach?

Because this affects multiple users, it indicates a central infrastructure fault. I first inspect the physical printer to verify it is powered on, has no hardware jams, and shows active link lights. Then, I ping its static IP address (**192.168.30.x**) from my console to check routing. If the ping succeeds but jobs are stuck, I log into the print server to restart the print spooler service. If the ping fails, I inspect the local switchport configuration and patch cables.

Q12: How do you stay updated with emerging IT technologies and security threats?

I maintain a continuous learning cycle by tracking reliable threat feeds like CISA alerts, BleepingComputer, and relevant security communities. I also run a personal hands-on home lab on GitHub to test automation tools and software deployments. Earning the Google IT Support Professional Certificate built my core technical baseline, and I am actively preparing for my CompTIA Network+ and Security+ certifications to ensure my skills keep pace with the shifting modern threat landscape.